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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
2nd Semester, May 2023

Paper Code: BS-112**Subject: Applied Mathematics-II****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Marks
1. (a)	Find the Principal Argument of $(1 + i)^{20}$.	(2½)
(b)	Find $\text{Re}(e^z)$.	(2)
(c)	Integrate $\text{Re}(z)$ along the straight line from 0 to $1 + 2i$.	(3)
(d)	Find the residue of $f(z) = \coth(z) / (z - i)$ at each of its poles.	(2½)
2. (a)	Find the Modulus and principal argument of $z = -1 - i\sqrt{3}$ and verify the result that multiplication by i is geometrically a counter clockwise rotation through $\pi / 2$ by graphing z and iz and the angle of rotation.	(5)
(b)	Find all Taylor and Laurent series of $f(z) = (-2z + 3) / (z^2 - 3z + 2)$ in the region $1 < z < 2$.	(5)
3. (a)	Find $\text{Re}(\cosh z)$ and all solutions of $\cosh z = 1$.	(5)
(b)	Evaluate $\oint_C [e^z / (z + 1)^2] dz$, along C where C is the circle $ z - 1 = 3$.	(5)
4. (a)	An electrical field is given by $f(z) = \phi(x, y) + i\psi(x, y)$ in the xy -plane. The potential function $\phi(x, y) = 3x^2y - y^3$ is given. Find the stream function $\psi(x, y)$ and the complex electric field $f(z)$.	(5)
(b)	Find the image of the infinite strip $0 < y < 1/2$ under the mapping $w = 1/z$.	(5)

****** End of Question Paper ******